ActInf GuestStream 125.1 ~ From Charles Darwin's "Root Brain" to Nikola

GuestStream | From Charles Darwin's to Nikola Tesla's ... | 1:20:03

Hello, welcome everyone. It is Active Inference Guest Stream 125.1 on December 17th, 2025.
And we are with a group of researchers who will each present in sequence, presenting on from
Charles Darwin's root brain to Nikola Tesla's 6G world brain and XAI native 6G networks.
So Martin, Nika, Osman, thank you all for joining and looking forward to your presentation.
To you, Martin. Thank you.
Thank you, Daniel. So thanks for having us. So this is a co-presentation of three speakers.
I will go first, followed by Nika and Osman.
Just let me see. Okay. So I will draw the big picture.
So the overarching goal is to realize an AGI, standing for Artificial General Intelligence,
native 6G world brain. Of course, using Active Inference, Nika will elaborate on the superiority
of Active Inference compared to Generative AI. And finally, Osman will show some ideas how to exploit
Active Inference for realizing explainable AI. So the vision of a world brain goes back to Tesla,
Nikola Tesla, who made this famous prophecy, when wireless is fully applied, the Earth will be
converted into a huge brain. This has been standing there for decades. The point is that with emerging
6G and beyond technologies, we are now close to realizing his prophecy. Enabling technologies such
as joint sensing and communication give rise to distributed sensing environment, which together
create this perception arcs, this sensing arc. Similar to Active Inference, there will be also a loopback,
an action arc. And there is a lot of research currently going on. I just mentioned two buzzwords.
It's about AI embodiment and agentic AI. So there needs to be research to be done. But once this is done,
you can see that this AGI native telecom brain creates a gigantic action perception loop similar to
Active Inference. This AGI native telecom brain has been proposed in this, in this, sorry about this
background noise. In this paper, you can find it on IEE explore. These are leading 6G scientists. The
interesting part is in this paper, the term active inference is not mentioned once. Likewise, Carl Fristen
is not mentioned once. These are stellar researchers but completely unaware of active inference. Of course,
we follow a different approach. We apply active inference and couple it with a number of other
strategies such as MIT's human AI strategy, which in a nutshell simply asks this question. So what would
happen if we replace AI neurons with people? Keep in mind, we want to build a gigantic world brain. And it's not
only humans, but also so called hyperintelligent cybernetic organisms which will emerge from existing AI systems
in the coming age of hyperintelligence. The good news is that these cybernetic organisms will not compete
with us humans, but live in a mutually beneficial symbiosis with us human beings. And because they are so
superior, they will consider us humans as we now regard plants. This sounds like bad news, but actually it's
an invitation for us to explore this so-called evolutionary continuum of human, and this is the official term
from ecologists, more than human intelligence. An example is the so-called Wood White Web. But this term,

Wood White Web, made some waves back in the 90s when it was published in Nature,

a more technical term is mycorrhizal networks. This goes back to two Greek terms.

Rhizal, that means roots, and myco, that means fungi. So these are two different species which live in

symbiosis. Symbiosis comes in a number of different flavors. It can be negative, such as parasitism,

as well as mycorrhizal networks. But mycorrhizal networks realize a mutually beneficial mutualism type of symbiosis.

So this symbiotic flora-fungi mutualism forms a new research field in biology. It's highlighted down here.

So it's all about decentralized exchange of information, resources, and exchanging reciprocal rewards to

create a community and resilience underground of all forests. And you have these roots underground,

and of course roots of different trees are separated, but there's a third player, these so-called fungi,

fungi that connect these roots of different trees of different species in order to create an ecosystem.

This has been observed years ago by Charles Darwin, and Charles Darwin coined the term root brain.

So this is a very packed figure. I mentioned it only some highlights. So remember, our goal is to build a

world brain. However, given that active inference is a biomimetic framework, we take inspiration from

nature, more specifically root brain. So we start with root brain, try to understand it, and then

Britt Britt

works properties such as transfer of resources from areas of abundance to scarcity from larger

plants to smaller plants so you see it's just the opposite of competition it's really collaboration

and mutual support and this is because of entropy entropy is the nature's hidden force driving life

to equilibrium in our work we also include a certain twist which makes things more interesting

as you might know from thermodynamics equity entropy must stay the same or increase but we use

generative ai to change the laws of physics by also reducing entropy this is done by using

generative ai more specifically we use generative ai to dynamically cobble two types of life cycles

so on one hand we have plant life cycles such as trees on the other side we have

mushroom life cycles so mushrooms are the fruiting body of those underground fungi so simply put we

we create cyber fungi which mimic fungi in order to hallucinate a certain type of network structure

the technical term is scale-free networks with small world properties in order to connect different plants

and their life cycles then these fungi might embody themselves into mushrooms and each mushroom has a

separate life cycle so you see here a plant produces spores uh it's pollen excuse me and mushrooms produce

spores the interesting part is now these are biological terms what does it mean in human terminology
so pollen might be translated into social norms like cross-pollination so we spread one pollen for
from one plant to another plant this is like in human societies we spread a social norm from person a to
person b and spores and a human equivalent of biological spores might be teleological tokens such as
purpose driven tokens widely used in blockchain networks so generally if i there are different types of
general api we opted for open eyes improve ddpm this is open source ddpm stands for denoising diffusion
probabilistic models and it comes in two complementary processes so we have a forward process called
diffusion this forward process increases entropy so it creates disorder and of course this backward
processes reverses the process by reducing uh entropy so creating order out of disorder this is illustrated
for a specific use case of ddpm image generation so this is the backward process we start with this order with
chaos technically speaking white noise and then we let the denoising process unfold itself and like like
magically out of thin air it hallucinates a realistic uh image which was not in the training data set so
this is really like like magic the trick and the challenge was to adapt ddpm from an image generation
context to our root brain context so simply put you have to understand how these fungi work
identify the processes and map these biological processes to ddpm's forward and backward process so
specifically the forward diffusion process of ddpm was used to biomimic the so-called branching process and
exploratory tendency of fungi conversely the backward denoising process of ddpm was used to biomimic another set of
processes carried out by the root brain namely fusing and homing process
this was all generative ai so far and of course we can do the same using active inference this is the graphical
model of our active inference uh generative model so you have a number of parameters those working in active
inference recognize them g is the expected free energy b is the transition matrix and this is the graphical
model behind every box of course you have a number of equations and as you know at least those working
on active inference a generative model is typically developed using this pom dp partially observable
mark of decision process this is what we did and then we were interested in comparing generative ai
to active inference for this we had to define a use case our use case was a bunch of people 64 people
humans and one person starts performing a new behavior and we were interested how long it takes
that until the new behavior is propagating through the entire population so we have two metrics i will
break it down on the following slides because this figure is a little bit too packed
important is how to read the x-axis you have to read it from left to right this is the forward process
of dtbm and then vice versa from right to left this is the secondary backward process of dtbm now i break it
down so we were interested in the first metric it's just the number of time units it takes till a new behavior
is propagated through the entire population and the black the blue solid lines this is the forward
process so we start with a number of curves this is dependent on a small parameter r this is simply the
spreading probability of the new behavior so the probability that a new behavior jumps from person over
to person b and as you can see at the forward process is unfolding the number of required time units is dropping

rapidly and then hits a plateau it stays flat and then from red to left we enter the backward process this is the dash red line and it's interesting what's happening at the end you can see the red curve stays flat it does not go back up as the
back up as the initial curves and this is very interesting because it tells us that with the
dtpm we achieve a minimum number of time units it takes to propagate a certain behavior through population
and the second important observation is it's completely independent of the spreading probability so it
doesn't matter whether you are more or less willing to adapt a new behavior generally if ai is so powerful
to keep the minimum uh at two time units then a second metric was kind of how to incentivize people
you know if you want you have a group of 64 people you ask them please adapt this and that behavior of
course this behavior should be desirable but you can also encourage incentivize people this is done by
token so whenever a couple of person people decide to adapt a new behavior each person is rewarded with
one token again i keep it short here due to time limit the green curve is the forward process these dash purple
uh curves are the backward process this bunch of curves curves hover over 2500 tokens roughly keep this number
in mind because we compared it to active inference so it's a typical figure it's like a cost function we
have to invest to to incentivize behavior the cost function is around 2500 tokens then we did the same use case
a bunch of 64 people with active inference first metric the time it takes to achieve full social contagion
it's a little bit slower instead of two time units it's one but it's comparable the main main difference is
in the cost function so instead of 200 2500 we have only 250 tokens to spend so active inference really outperforms
generative ai by one order of magnitude by a factor of 10. why is it because active inference as opposed to
generative ai is able of life long learning this is illustrated here on the y-axis we have the belief
update magnitude and on the x-axis the active inference process unfolding so let's start with dtpm the
generative ai this is the dashed yellowish curve you see there's one learning event but then at the
process is unfolding no more further learning is taking place very different from generative ai active
inference there's a huge learning peak at the beginning and then there is continuous learning
throughout the process so one question we ask ourselves is whether there's a way to further enrich this
lifelong learning process of active inference and this brought us to a new field it's ai and psychedelic
medicine believe it or not this is a nascent intersecting area of generative ai and psychedelics
psychedelics is now part of mainstream medicine to cope with mental health issues such as depression
and anxiety and anxiety and anxiety and the trick is of course we want to make ai not only a biomimetics
but also psychedelic mimetics meaning all the benefits of psychedelics are also realized in ai of course the
question is how to do this so the main challenge is how to integrate the cognitive mechanisms activated by
psychedelics into our ai design doing this also has the nice benefit of addressing one of the shortcomings
of active inference as you might know active inference is all about surprise minimization this necessarily
leads to this well-known problem of dark room problem so if an active inference agent continuously
performs active inference meaning surprising uh minimizing surprise the agent will sooner or later end up in

a dark room because nothing is happening there no surprise is happening there of course this is counter
to what we expect from life but instead of seeing it as a problem we should see it as an invitation to close
our eyes and turn our attention from the external action perception loop inward okay and neuroscientists use this term
brain's dark energy so whenever we close our eyes our brain is not doing less just the opposite some parts of
the brain really lit up and consume up to 20 times more energy when at rest and this is the path forward so we
have to extend expand active inference by a secondary complementary uh theory it's known as entropic brain theory of course
karl fristen already worked on this so psychedelics have two benefits it not only widens the free energy
landscape but it also flattens the minima so it's easier to escape a local minima in order to explore the widened
uh free energy landscape so i'm almost done and then i hand it over to nika so in our work and we can report
some results on this our aim first aim was to advance active inference from a biological aging context to a
psychoactive agent luckily nature produces psychoactive agents which come in three flavors stimulants such as
caffeine sedative if you undergo a surgery you use sedatives such as opium and of course the third one
hallucinogen such as psilocybin and the interesting part is only this the third one produces this
characteristic this feature we are trying to harness mental plasticity so only hallucinogenic agents are
capable of inducing the state of mental plasticity and this is what we want to mimic in our ai design
and to do so we had to go back to the generative model of active inference recall typically it uses
the concept of pomdp so it comes with a standard set of variables and matrices it not none of them is
suitable to accommodate psychedelics so you have to explore the so-called pomdp pomdp extension and then
you find something which captures the prior belief of agents d and our goal is to induce mental plasticity
and to do so we have to weaken uh the prior belief d of active inference agent and this is done by
by decreasing a certain parameter known as precision parameter omega norm and this is my final slide
so this is something just for illustration how ebt really kicks in on the y-axis you have the entropy of
prior belief okay and on the x-axis again the active inference process unfolding we compare two different
active inference agents the red one is a conventional active inference without any ebt enhancement and it
does what it's supposed to do throughout the process the agent tries to minimize surprise meaning
minimizing entropy and this is for survival it's good purpose for learning it's not the best strategy
because minimizing surprise of course avoids surprise and epistemic opportunities the second agent was
put under psychedelic for a temporary window between time step 500 and time step 1000 then you can see the
entropy of the second second agent's belief really shoots up and then it stays flat this being flat also
nicely um reflects this so-called uh psychedelic afterglow effect of psychedelics this is good news because it shows that ebt is really able to increase entropy
and increasing entropy in our terms means you know it enables divergent thinking and this is it instrumental

in in uh increasing creativity of our ai agents and 6g world brain

and this is all from my side now i hand it over to nika thank you

okay i will now share my slides

okay so hello everyone my name is nika i will now go into the implementation details of both generative ai and active inference agents

uh compare their difference and compare their performance through results and then explain how we actually modified it with sleep rest and psychedelics to solve the dark room problem

okay uh here we compare generative ai versus active inference with generative ai we have

this embodied passive ai but with active inference we have embodied active ai

embodied means it has an actual action perception loop that acts in the word with generative ai we have no belief updating after initial

training but with active inference we have continuous bayesian belief updating

uh here outputs are created with generative ai without acting in the environment but with active inference

actions are selected by uh inferring posterior over policies with generative ai we have pattern matching

perception it means that generative ai does not truly understand inputs it just detects statistical patterns in

the data and just matches the statistical patterns but with active inference we have perception as inferences of

latent causes it means that active inference actually understand the hidden causes behind everything that happens in the world

here we get inspired by the mycorrhizal networks so mycorrhizal networks has three features that we get inspired by

branching fusing and entreating with branching we have exploration so fungal filaments extend into the soil which may make outward

growth as exploration and increasing entropy and fusing is kind of an exploitation so connection fused to redistribute nutrients and

restore order and resource consolidation and with entreating or embodiment fruiting bodies emerge emerge to act in the world so it bridges

the biological and biological um digital and biological agency

here we created 1.3 million small world scale free graph structures using network x in python so using network x we created

these graphs small word means the network has tightly connected local groups and scale free means that

the network x in python so using network x in python so using network x in python so using network x in python

and many smaller nodes so we here we encode the symmetric binary tensors at zero and one so we have 64 humans

that these humans are representing in as nodes and the social links are representing as edges

so here when we have an edge we put one when we don't have an edge we put zero and we feed this data set into

um the ddpm so we get adjacency tensors at outputs but in output we also necessarily don't have the binary output so

we have a two-stage binarization threshold which is 0.5 if it's above this threshold we put one it means humans are
connected if it's under this threshold we put zero and here some cyber fungi acts as an adaptive generative
ai agent and we do dynamic edge rewiring with teleological tokens so here we try to do a social
contagion norm so the spreading probability is between zero and one so this is a general overview
of the pomdp based graphical model of active inference i will now break down the slides into detail
so the agent model components are state factors control factors and observation modalities so
here state factors are snorm smb snet and stoke s norm is better agent adopted the social norm and smb is for
meta beliefs so is what i think my neighbor beliefs and snet is the network structure which means presence
or absence of edges and stoke is whether cyber fungi issued a teleological token for control factors we
have cyber fungi agency and human agency with cyber fungi agency we have unit and utoke unit is reviring
adding adding adding or pruning edges to accelerate contagion and utoke is token assignment to reward norm
adopters with human agency so humans attend to norm and meta belief because humans should decide whether to
adopt or reject the norm and they should decide whether to select a neighbor to sample new evidence from and
also observation modalities are the same we have four categories and humans attend to
on norm and onb and cyber fungi attend to onet and auto here are the generative model parameters so we
have a likelihood tensor which map hidden states to observation probabilities we have a transition matrix
with show temporal evolution of beliefs and priors d and e so d is initial beliefs over hidden states and e is
initial beliefs over actions and there is a c which is the outcome preference vector which is the goal vector that we have
this slide summarizes the free energy principle as a unified framework so pre free energy principle
measures the difference between what the agent believes and what's true and each agent minimizes f to
reduce surprise and select optimal action and expected free energy is about the future so
we try to anticipate the value of future outcomes under the policy p with this formula
a key challenge with active inference is that we have computational attractability because we have 64 humans
the size of transition or likelihood matrices is 2 to the power of 64 times 2 to the power of 64.
how we solve this challenge so we divide these 64 into each sub factors so
these sub factors have either base plus 1 or base so then only we need a 2 to the power of bi times 2 to the power
of bi likelihood over transition matrices
so this slide compares ddpm with active inference so well with r equals to 1 it means spreading probability
equals to 1 we see that aif time is 1 and ddpm time is 2.5 for social contagion of all nodes
so not only active inference has less needs less time to have the social contagion of all nodes but
we have 10 times more token efficiency so aif achieves order of magnitude lower token cost than ddpm

this slide looks at the small word index so the small word index this is the formula and it shows the ratio
of clustering and path length relative to a random network a rule of thumb says if the sigma or a small
word network is bigger than one we have a small for regime here if you look the ddpm agents small word
um a small word index is much more higher than the aif agent so ddpm preserves a small word structure
better than aif here's why because ddpm shortens the path length and also increases the clustering
coefficient here is the belief dynamics so delta kl is obtained with this formula and it measures how much
an agent updates its belief from one step to the next so it's kind of a kale divergence of qs and qs minus
one here aif has multiple recurring spice uh spikes during um which are decreasing over time which shows
genuine learning and ddpm only has a single early uh peak which drops to zero which shows the minimal
learning here so professor meyer completely explained why what is darkroom problem now i'm gonna explain
what is our resolution so the resolution is that it's not just the brain minimizing uh the surprise it's
the agent environment so it means that uh the resolution is that uh in stimulating environments
the best way to reduce long-term uncertainty is actually to explore this is called the enlightened room
view and here we have the neural evidence which is the break brain's dark energy so brain uses 60 to 80 percent
energy address and the default mode network or the dmn is active in rest for simulation recall and press
preparation and intrinsic brain activity is 20 times more than external stimulus response so it highlights
exploration exploration exploitation as a baseline so we try to disengage from action perception loop through
strip rest and psychedelics to allow exploration within internal space here is the first modification
we do through sleep cycle which is the bayesian model reduction our core concept is that bayesian model
reduction prunes redundant connections similar to the sleep cycle and in absence of new data complexity
minimization becomes our first goal here we calculate delta f which is the free energy difference between two models
here you can see a is the posterior concentration parameter from observed data and a tilde is the prior parameters of
the full model a prime is the edge removal candidate for model reduction and b quantifies the model evidence
so we calculate delta f from here and if it is less than zero it is in favor of the reduced model
there are two a stage for sleep routine stage one is resetting priors for each hidden state factors
so we override prior df to posterior qf and this naturally removes the kale divergence term between qf and df
the second stage is pruning edges so for each edge u and v we construct a reduced model and compute the delta f
if this is less than negative three it means that we prune the edge this was the original bmr so we have we
propose an enhanced bmr cycle which has repeating cycles of wake wake and sleep this is a loop which shows
continuous adaptation in during each wake we add three edges and during each sleep we remove the weakest 20
percent of posterior qnet edges here are the results of the beijian model reduction and confidence dynamics
so here you can see that we have the single cycle bmr that f reduces from 10 to 9.75 which delta if is 0.25 nets

here is the multi-cycle with guided pruning so it is our proposed bmr you can see that it goes from 10.5 to 9.25 here so it reduces by 1.25 nets so it has a five times larger reduction in free energy the next metric is confidence dynamics so confidence uh is showing for example high confidence is clear agent rule confidence is obtained from negative entropy of qnet so and lower confidence is showing a structural uncertainty or uniform q here you can see that the multi-cycle agent has higher confidence than the wake-only agent

the second modification we do is through rest so during rest the brain exhibits spontaneous activity across resting state networks and rsm maximize uh spatial templar entropy reflecting the brain's generative model for future adaptability and accuracy is the emphasized here so we want to map rest phases to pmdp based a framework we have five stages first we disengage from external inputs then we suppress accuracy and try to do maximum entropy policy sampling so we try to build the fictive trajectories and sample from categorical priors and categorical transition matrices and using these frictive trajectories we try to update the durichsthal parameters a and b

here are the results we build a spatial a spatial entropy temporal entropy and a spatial temporal entropy here spatial entropy measures the diversity of network clustering so here you can see that wake plus stress agent has more spatial entropy it means that it has more distributed and flexible connectivity with temporal entropy we we capture irregularity of norm switching here you see that there is a little or no difference between them which higher um temporal entropy indicates more asynchronous norm switching between agents and spatial temporal entropy capture both spatial and temporal

entropy features so you can see it's with wake plus rest agents we have a little more spatial temporal entropy and because we suppress accuracy in the wake plus rest agent you see that accuracy here is less than the accuracy in wake only accuracy shows the negative expected surprise and it's the predictive alignment between beliefs and observations

the third modification we do is brain theory unification under psychedelics so professor fristen's work merges free energy principle with entropic brain theory which links computational neurobiological and experimental insight into a single explanatory model

here we try to do precision relaxation so using psychedelics we relax the precision of high level priors this fosters openness and cognitive flexibility then we then this increases entropy so psychedelics temporarily increase brain entropy which allows escape from repetitive patterns of thoughts and promoting a novel neural trajectory

and then we have expanded access to the neural trajectory and then we have expanded accessibilities so this expands the brain's accessible stage so it allows the emergence of new cognitive configurations

here we deactivate the default mode network the default mode network or dmn is the part of the brain responsible for self-referential thinking and rigid patterns of thoughts and then we have the psychedelic

psychedelic afterglow which is the period after psychedelic effect wears off so the brain settles into a state that is stable but more flexible than before

here are the results of cognitive dynamics under psychedelics which shows the prior entropy here you have the prior entropy results the prior entropy measures the uncertainty of priors here you can see that with psychedelics and through psychedelic window

we are we have more prior entropy we are we have more prior entropy which means we have flexible priors and it also stays the same after the psychedelic window which means we have the psychedelic afterglow effect

Next we try to second we try to um show the posterior synchrony uh so we measure posterior synchrony or belief coherence which captures how aligned agent beliefs are across the networks so here high synchrony means that we have coherent beliefs so with wake only

agent we have coherent beliefs but with wake plus uh with a psychedelic agent we don't have coherent

beliefs so it's it's trying to enhance the exploration instead of coherent beliefs

our final use case is the comparison of aif enhancements so we compare all the prior

entropies of all agents with sleep rest psychedelics you can see that sleep has most

prior entropy and wake has the least prior entropy

okay here osman will take over and explain about active inference and explain explainable ai thank

you thank you nico let me share my uh slide set oops so you can see i guess right now yeah okay

so uh first of all welcome everyone i'm osman tugai uh on the behalf of technical university of berlin

and fran hofer hhi institute but we were collaborating with inrs especially of course professor meyer last

one year over and i was visiting him this summer this summer in the canada so that's i will try to do

um basically come up with the um how we can combine active inference in the in the in the sense of uh

next generation wireless networks specifically i will uh after this theoretical size of the research i

will come up with the more application domain for the next generation wires as i said 6g and 6g beyond uh

uh architecture and my like my research focus was explainable ai and i i'm an ai scientist for

wireless so i will try to combine these two methodology in a way that with our recent research

research so let's start with the uh our very foundational information age um um um terms and

the paradigms which is the uh maybe in in your community it's not maybe very familiar but like

cloud shannon was the like define the nowadays that we are knowing what is the communication network so

how the one two asset can communicate in a way that information source to destination source how the

message goes basically the how today whatever you know about the wireless or telephone technology is

comes comes up with this diagram the basically the we call it this information age that was the

end of 40s early 50s this this design was happen but um while he was uh designing information age

he was also of course interested on the learning machines thinking machines as like alan turing but

while he was doing this specifically as our today's um talks uh the the title darwin's root brain

theory he was also aware that darwin was also um coming up for the like this uh bio biology inspired

machines so cloud shannon was also was aware of it while designing this wireless um um

um the communication backbone he was also working on the learning learnable or like machine uh

machine technologies or like logical machines so this is important why because like last neuro ips

conference i was talking with the uh mihaile van der schaar from university cambridge so the idea behind

this wireless community and the ai community needs to work back to back to design uh efficiently of this

the the the the upcoming uh technologies because nowadays 62 percent of the worldwide internet is used by mobile networks while this ai interaction happening so that's very important actually the statistic when we're thinking about it okay after the very theoretical introduction then when we check about like the we call it nowadays what is happening on the communication is we call this this this um period is um post channel era post channel era basically what is as you can previously for the first slide you can remember that what we are like trying to send the raw bits towards uh transmitter to receiver uh it's classical like channel channel is our the you know the uh the place that where we send over to our uh message so but you can see that post channel is a bit different because now we are started to train our ai models especially the artificial general intelligence models like foundation models to make more uh efficient coding um network coding on the like physical layer or if you go i will of course come up with new slides and regarding our uh research it's even for the upper layers like the semantic knowledge so rather than to sending only raw bits like you know the data it can be voiced can be whatever the image now we can basically tell the network that what is our intent what we want to send it to receiver so that's the nowadays we call it post channel era that's why the ai is heavily integrated on the 5g then we are trying to make uh basically the ai native 6g so that was the reason and you can see from here as i said ai already on the 5g is deployed so many uh different countries different factory settings or self-driving cars anywhere because when we're thinking about today's like why more in the like san francisco this cars needs to communicate with the edge servers over the wireless so that and when we're thinking about use cases ullc means ultra reliable low latency communication or the embb like the video recording or video screening should be very like prioritized on the network this can be a prioritization or the resource allocation can done with the conventional ai techniques but this is not enough because we are going further more subject sub 3 gigahertz activated bands which means more data rate and more capacity of the network then we can basically go more uh it's it's called advanced ultra reliable low latency communication so this is coming towards near real time interactions with the network towards the uh components of the real real life so how it looks like i mean when we're thinking about we're starting early age some of some of us maybe catch 3g some of us maybe 2g and do you realize it how it's changed basically you can see that what we are transferring was text voice now we are talking about space communications tactile internet like the today's like like like optimus robot hand um and interactions with the real time um uh life in the like or the act of life and as well as automated cars like uh um like drones or digital tvins there are the they are the now the the data sources or like the data um applications that we are using towards 6g so that's very important and as you can see we are like 50 kilobits per second to one terabit uh terabyte per second communication we are talking and included quantum communication we call it like quantum aware communications in the in our domain so this is a this is a big change basically it's not like from 2 to 3g so we are basically designing the ai native uh ai native 6g so every layer of the protocol stack is communicating with different sort of um ai algorithms or let's call it learning learning um learning algorithms because some of them can be conventional uh you know machine learning techniques because you don't need to much more reasoning capacity or like cognitive inspired agents there but most of the time uh every layer it's this now we are designing to deploy 2030 to this technology

with ai native so that's very it brings very important responsibility that's why one of the
very important pillars of this um ai native 6g design is secure and transvertence so why this is very
important you can see this the itu is international telecommunication union and 3gpp is the nowadays like
any of generation of the technology like 2g 3g 4g 5g is defined the specification of the 3gpp which is
included different companies different institutes they are like the biggest institutes in the world that
they define the mobile or wireless networks uh specifications so while they are doing this when
they when you check their recent reports i2 our imt uh like the 2030 report as well as the 3gpp is the last
specification one of the biggest important uh pillars are trustworthiness trustworthiness means
since we are designing ai native 6g we should aware that there should there shouldn't be like a problem
regarding trustworthiness explain or interoperable um decision points in the network uh while the
network operating because there are so many um communication environments like energy grid networks
like you can think about you know this the electricity problems happen in portuguese or like this the
nuclear central or like you can think about teleoperations smart hospitals or very low latency
applications on the smart factories is like a very high production lines so then this means that while
okay we are doing this ai native design we should be aware that this ai doesn't degradate the network
performance so that's why this trustworthiness or explainability aspect of design come to play now i
want to show that okay we are we talk about coming from the ai native design to explainable aspects
and i want to come up with our uh in uh technical university of berlin what we did some explainable
ai research and then how we like we can um get advantage of the active imparance in our use cases
one of the use case for instance remote controlled robot you can use this robot anywhere like it can be
some earthquakes situation it can be some exploration purposes anything else so while you are using that you
can maybe want to allocate your resource because there should maybe we use in the scenario one one real
real robot remote controlled robot but you can use so many of them so then you need to be aware of your
network uh usage or network uh resource allocation problem that's why you can use some prediction
models on the top of your network it's called that's why ai native and while this network communicating
it can be op you can your ai algorithm can operate some uh resource allocation problem for your
components of your design but again very important we are now talking about earthquake there could be
some lives or it we can talk about this robot can be basically working on the smart factory environment
and making some operations so then we should be aware again our prediction ai model or like learning
model or like deep learning model or any reinforcement learning agent should be transparent
and not black box that's why the we designed in our ai algorithm top of it some xii design to make it
transparently available any other any other story other story this was like a work with the northeastern
university from us as well as india networks from spain and and of nsf pondered and we i mean one of the
very biggest topic also in our communication or like any other even when you check the companies digital
events which means very low latency applications of the real world components to uh digital replica of
it why then we can basically simulate so many different scenarios um of like that real life component
that could be production lines what could be again like the smart medical applications what could be

robotic environments nowadays when we are talking about this humanoids uh their digital tvs can be trained in a
low cost and more efficient domains or even i was at neuro ips and there was a tesla's presentation
they are basically training their uh controller uh for the like this uh so-called digital tvn simulator so
that type of applications are very important so you don't need to deploy the whole bench of network or
technology stack so that's why we made it in our collaboration we got the net one of the collestium
uh digital tvn to create a 6g network or 5g network in this sense here and then we basically created
some different sparse datas with the generative ai so the the main idea now we are working on that to
make okay we are creating some synthetic data for the like digital tvn layer but is it really performance
okay in a in a sense of ai kpis it's it seems good but we should check that is the model is uh
operating in a good way that's why the xai was coming play and and one another uh story was again
in the molecular communication domain we in our projects called internet of bio nano things so this
is also very important for the active inference people as well as the nikas and the professor meyer's
side of the theory because now it's basically if this ai models work for the biology um biology um
environments so we were basically what we did is we again when you think about the communication
network it was immune cell was the um uh transmitter and tumor cell was the receiver because when the
tumor cell comes towards the immune cell immune cell starts to release some sort of uh um biochemical um
components so that's why we try to use some ai algorithms like especially very simple uh uh
feed neural network to understand that which uh there was a two feature one is the like the estimate the
distance another one was the released amount of molecules so while we using this ai we want to make
it which component is much more efficient in our this nanoscale design because it's very important then
we have a knowledge about the okay this feature is more informative in the sense of cancer research so
that's why we use explainability in the sense of this research lately i said the in communication there
there should be there are some research domains on the physical layer but as well as the our latest
research with the like big collaborative work we try to basically on the higher level of the network
components semantically we want to transfer the knowledge i don't want to get that much detail
but in this work we try to basically get um reasoning capacity and then the human expert can communicate
with basically the network components in a way that with llm way like the gpt models foundation models so
that was the reason that's why we call it reasoning model so whatever it happens in the lower layer like
let's say physical layer parts of the network it gets it generates some graph neural uh some graphs and
we convert it to knowledge graphs basically the human in human interpretable statements then human can
interact with the like network which is normally it's not the case in our today's conventional
communication mobile network so that was the novelty of this work so uh when we're thinking about this
is like this is a big change i'd say that starting from shannon to post shannon this was the from
huawei's report and they also like in the globecom they presented efficiently they want to change the full
like nowadays have we knowing the the communication networks they want to change it for the like full lm gpt
wise network for instance they call their user equipment which is our phones is uh for them um
uh a iphone or or like they call it ai ran so everything is work with the lm uh integrated agent

equation so which is a which is upcoming reality so when we think about like it's again uh this this this movement towards agentic design for so many different applications as again professor mayer and the nikah was uh focusing on the uh in their talk like 3d networking blockchain technologies or holographic telepresence we should also like as we said as a wireless designer or like the 6g scientists we should come up with more um um cognitively capable agents so that's why the point of active inference was coming to play with our discussions with the professor mayer and then we discussed like how we can do it and then my interest was more explainability and he he was on the like the more design of active inference for 6g so we started of course with the state of the art so what is happening in our in our domain what's uh what is like usage of the learning techniques most common technique in everywhere again reinforcement learning techniques for we call it network slicing so the network component network resources sliced between the prioritization of the scenario in this sense maybe uh autonomous car very low latency communication some maybe smart hospital or maybe iot sensors or maybe some ar vr glasses so for them that's an efficient work for the like uh nowadays they use it in the wireless domain so this is again traditional way and then we designed in our like well-known uh backbone of the um active inference agent we call it brain because this is the from the our paper's name based in reasoning where active inference for explainable mobile networks why we did it the idea is again the same comes with this trustworthiness you can use it reinforcement learning but you are not sure agent's behavior it can be a degraded network performance but we can use the posterior beliefs here to represent the agent's beliefs towards the upcoming decision action point regarding the states that's why we come up with this design i'm though i don't want to get that much deep details because it's a very long paper and under the review of conference and the journal so that's why i want to give up like what what design in a like a bit abstract way so what we did is uh more than theory basically we applied uh active inference uh design brain agent to directly a um gpu accelerated radio access network which in the sense of different users three different users and then we try to basically visualize the agent's beliefs with the help of this the posterior belief design of the active inference itself by nature so this is inherently there so that's why we want to use it in a way that how agent is seeing the network uh network um state so as you can see for instance in our three different slice scenario in in embb slice which is like this most like most connectivity like you can think about arvr application here agents start with like he say that like regarding the first initial data prioritization uh in initialization he the agent or he she uh or it uh directly says that oh there's a high demand which is actually when when you think about it it's it's it makes sense because the there is a demand of the uh down lake from the application domain so that was makes sense and this was our checkpoints and in some some sort of uh after the uh the user equipment reached the network the agent itself it says that okay now it's getting more lower and this is like uh as you can see this is the like belief probability the belief probability now the demand of the demand state or like the need of district resources for embb slice is lower and i mean same for the like different i mean different priorities or different interest or different demand state for ultra reliable low latency but we were like efficiently see that agent's behavior in a posterior belief state of course in the paper uh there is much more extreme and the instructed uh agent uh behavior and then it's it's it's a it's a long paper so i would be happy to share uh when the results came so um lately again very fruitful result for us was the like when we check about the mean cumulative reward we because we want to in our scenario we want to see the conventional reinforcement learning techniques success and compare with these baselines and you can see that our uh the the brain agent basically much more efficient in our non-stationary wireless environment than the well-known uh like you know actor critic or deep crew network or like policy gradient network is much more efficient than the average and because that some of them is gets sometimes the plateau and goes up and it's it's it's classical behaviors of reinforcement learning and then the the lost function was more efficiently stable in our scenario and here i'm finalizing my talk thanks for listening for us now i think we can get some questions questions thank you awesome okay yeah martin comments from any of the other authors but while i get everything back going yeah we just were discussing among us while you were gone i think our talk is kind of overwhelming nicely put or uh more i know it's to some and it might be even frightening you know not because the topic itself you know but it's just the ground we are covering you know we go from charles darwin from nikola tesla it's she application so it's just i think it's well nicely put there are many many things you could could ask question about on the other hand you know you might say so what is the major the d1 takeaway you know you know so so danny i feel for you you know i feel sorry you know uh yes i do have a few i guess entry points um especially for the the active inference people who will be familiar with some of the ways that the palm dps have been presented so i guess just to begin with in in the cyber fungal network what cognitive functions does it perform and what kinds of transfer would you expect or how do you adapt it to other kinds of functionalities like the kinds of things that that we can use language models or other methods for today so i would i would you know yet understand it's like the brain you know so you have the different cognitive functions such as perception memory it's also a goal directed behavior you know so these fungal networks pursue a certain goal you know it's kind of to build a an ecosystem a community for exchanging not only information but also tangible nutrients you know resources so it's really it's everything active inference framework is so what we're doing is incorporating such as goal directed behavior memory action perception is found in the in in the uh root brain the interesting part for us is more kind of okay we we don't want to build a little ai or a little human brain this fungal networks happens to be the largest living organism on planet earth so our goal was to build a 6g world brain so our motivation is then let's learn from the largest living organisms on planet earth which happen to be this fungal network and we are not at the end of our question so it's something it's so it's promising where to go next i think it's it's really this what what is currently missing in 6g research this uh ai embodiment and agentic ai so these fungal networks are underground in nature but you might remember i also said there are mushrooms so mushrooms are the fruiting bodies of fungi so put differently the mushrooms help fungi to embody the space above ground ground so mushrooms are like embodied fungi so this is one type of so mimicking fungi for us gives some ideas how to realize ai embodiment and these mushrooms also engage in certain activities so mimicking again

mushrooms might give some some hints uh at realizing agentic ai so this is from my side i i i have to say

a large empath

whether those were observations or whether the tokens were embedded,

like they had already been passed through an embedding model of a LLM.

So I handed over to Nika to give her the opportunity to...

Thank you.

So here, when we do this with DDPM, okay,

it's like some tokens when we have the...

So this is at each step, okay?

For 4000 diffusion step, we are creating every DDPM graph structure.

So we have 64 humans.

As this connect together, we have a link or not.

If they have a link, okay, if there is a link between two humans,

there is a token as a reward assigned to either of the humans.

So one token to one human at the end of one link

and one token to another human at the end of the other link.

This happens the same exactly with the active inference.

So here we can say actually the network that active inference creates

has much more, less edges than the network that DDPM creates.

This is the meaning of token assignment.

So it has much less edges, but it has the same efficiency.

Why?

Because the time it takes for social contagion of all norms is the same, basically.

And even for spreading probability, which equals to one or 100%,
it means that when some human has the social norm,
the other one that has a link to it, 100% adopts the norm.
So it means that with, in this condition, which are equals to one,
AAF time for social contagion of all nodes is even less than DDPM time.
How much less?
AIF takes one time unit and DDPM takes 2.5 time units.
Yeah.
Okay.
So the tokens are being, they're kind of a, it's like a currency within the simulation
that's being passed amongst the agents.
Okay.
Yeah.
So, and you're calling the nodes humans and were they people or just simulated agents?
No, they're agents, active inference agents in active inference and nodes in
DDPM.
Okay.
What was the teleological token?
Teleological token is kind of a reward based token.
Okay.
It's a purpose driven token.
So what is our purpose here is to, as fast as we can, we have to do the social contagion
of all nodes.
So all of the nodes in the network, which is 64 humans have to adopt the norm.
So we want to see how fast and how efficient is that.
And the token is like a cost for us.
Okay.
Why this is a cost?
Because we cannot have like as many as edges that we want.
Okay.
But with so much less edges, we can have the same level of efficiency in active inference.
So like, this is much better.
Okay.
So how do you go from what that network does to the kinds of tasks that people are working
on or the kinds of ways that people would want memory or perception and action to be
working?
Okay.
So for, for example, perception and action, we have agent model components.

So we have some hidden state factors for active inference.

And these hidden state factors, then we have agency.

So we have actions.

So because active inference have to have an action perception loop.

Yeah.

So they have to act in the world.

So what does cyber friend fungi do?

It actually assigns tokens to each of them.

So as I said, it assigns tokens to each of the humans that has are at the end of the edges.

So when there is an edge, we have two tokens for each human.

So cyber fungi does this and also does the rewiring.

So for example, cyber fungi can remove one edge and add one another edge.

How does the cyber fungi do this based on free energy principle?

So there are candidates, for example, we want to add three edges.

Okay.

There are candidates and then cyber fungi calculates the free energy.

What happens if we add this edge?

What happens to the free energy of the network?

And it only if it reduces the free energy, we can add that edge.

If not, we don't add that edge.

So the policy is leading us toward reducing free energy, which is based on free energy principle by Carl Friston.

Yeah.

So how would it memorize or operate on like a text string?

Right now, there's one kind of token being passed around these different agents.

And then these network topological model selection driven operations.

So would it stay with one kind of, um, let Martin back in.

Would you expect it to stay with one kind of token or would there be multiple kinds of tokens to support memory?

A bit active inference.

We actually don't have memory because, uh, like this is randomly some candidates of edges are selected.

Then we have the free energy.

So we have to calculate the free energy and go whether this reduces the free energy over 4000 steps.

So because we have 4000 steps, it is done like as we want.

Uh, and also, uh, what was the other part of the question?

Sorry.

Yeah, definitely.

it has the training data set. So it trains on 1.3 million networks, small, a small word, scale free networks. And at the end, it's trying to create the same features, the small word feature and the scale free features, it creates it, but like with much more token cost. And also, what was the other part of the question? Sorry.

Whether you would sort of scale or develop this with one kind of internal network currency or multiple different types? Yeah, we can implement, we can definitely implement different kind of tokens, each of them meaning something. For example, we define tokens as whether there is an edge between two humans and we assign two tokens to each of them. But we can, because this is a purpose-driven token, we can assign different tokens. So this is based on us, like we can extend our work in this direction, definitely. Yeah.

And is that, just remind what paper and is that code available for that first part?

The data set is available online, but I haven't uploaded the code yet. And we only submitted the paper and we are waiting for the results, but I will upload the code soon.

Cool. Yeah. What are the next moves? Like there's definitely a lot of sides to the development of your research, but what do you feel like your next moves are?

Yeah. Because we, we did some kind of embodiment here. So we did embodiment with DDPM because of some action that we did on it, but we have, we want to push this burden. We want to move forward with this embodiment. So as we said, we're trying to move from the fungal networks that are underground to mushrooms.

Now we want to get to the above word. So mushrooms are the fruiting bodies or embodiment of the fungal networks. So we want to move to some kind of 3D spatial internet, which we want to implement it with the holographic content to helps us with the, for example, any use case that we want with the social contagion, we can do it, or we want to like have a real time 3D spatial internet. We are currently working on that.

So, Osman or Martin want to add any sort of closing comments?

I want to, I want to hint at, to Osman, I think this explainability because active inference is very, very suitable for enabling explainability. I, maybe Osman, you can make some comments on.

Of course, of course, of course, Martin, as, as I said, I mean, I give a bit sneak peek of results.

So what our, I mean, most of the, the focus was the nature of this posterior beliefs. It's very important.

It's very informative side of this, the, the, the, the theory of active inference, as well as this, the free energy minimization. So in that sense, we can get so many informative results in the scenario of non-stationary wireless data domains, like, as well as, as I said, radio access networks. So that's at the moment, our, this two extended work is one on the conference and then the paper, but there are also still so, because it's very new in our domain. I mean, we, we can get, of course, we checked now, as I said, the, the agent's station of awareness, but we can also check the action and the minimization of the free energy principle, and the, let's say, interpretability. Then we can get more information about the agent's behavior, which is very informative in the sense of, again, very critical networks, or low

latency applications, as again, one of the low latency applications, as the Nika said, is a holographic communications, or like, let's say 3D internet. In this type of settings, when you want to use this type of AI agents, it would be the, like, the one of the, like, need and constraint will be the trustworthiness. So this will be our extended works towards different use case. One, another side I want to definitely want to explore is the smart healthcare, or like, let's, let's say, like, this type of application, because they are very vital. So you need to be very reliable in these networks, even if you are implementing some AI algorithms. Yeah, definitely, there's a lot of, a lot of work there with the explainability and the post-Shannon semantic information transfer, in a way, is like, explainability, like, distilling some world processes to their causal basis is the accounting. Just taking a higher resolution video is more syntactic information, but it doesn't necessarily convey any further explainability.
So when, like you pointed out, there's that cognitive agent to agent transfer, just increasing the bandwidth alone doesn't capture or necessarily correlate with the semantics of the communication. So definitely a big emerging theme. And because there's going to be these different agents of different embodiments, you don't necessarily have one template, which is kind of the importance of having a unified cognitive approach, whether the communication is to, I guess, a chitin based or digital fungus, as it were.
Also, lately, I want slow, like, I didn't, again, like small edition about this explainability side of story, you know, also because of the our, like interpretation of the free energy, we can basically get again, like, if you want to like, make it more interpretable, a free energy minimization story in the like, let's call it the agent's behavior. We also checked in our paper like this, the so called, like preference score information gain. So this is important, because then we can also like, understand what is the tendency of the agent that wants to do in this sense of scenario.
But that was also the one of the our, like, let's say experiments in the paper. So I'm, we call it extrinsic and epistemic sort of exploitation. So that's why, I mean, I'm, if the results are coming, I'm happy to share. And maybe the, I was checking today to maybe coming to next active inference workshop, and then to be in, you know, interact with the community about this works.
Thank you. Cool. Yeah, definitely. I know many people will be interested to see the paper in the code when they arise. Any closing comments that either you would like to make?
Oh, thank you, Daniel, for hosting this meeting. And thank you for your time and attention and your questions. We really appreciate this. Thank you so much for your time, Daniel. Nice to meet you again.
Yes. So good luck. Happy travels. Looking forward to the next steps in the research.
Okay, thank you. Have a good day.
Thank you.